

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs (2150 BCE – 2026 CE)

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Abstract.

We analyse a merged catalog (4,418 CSV records; 4,267 $M \geq 6.5$ events; 2150 BCE–2026 CE) using the Baiesi–Paczuski metric η with tectonic-path distance (Bird 2003). 47 global seismic series are identified (27 modern, 15 early, 5 historical candidates). Significance: permutation test ($n=10,000$, $p<0.0001$, $z=-6.17$); ETAS validation ($FPR=0/100$); FDR (45/47 at $q=0.05$). Fifteen early series reach $p=0.007$, but pre-1960 incompleteness limits interpretation. No historical series are significant ($p=0.46$). Largest modern series: S170 (46 events, 12 Flinn–Engdahl regions, $M_{max}=9.1$, 2002–2023).

Keywords: global seismicity; seismic series; earthquake clustering; tectonic distance; Baiesi–Paczuski metric; ETAS validation; false discovery rate; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that $M \geq 7$ clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global $M \geq 7$ and $M \geq 8$ rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. We test the hypothesis that multi-regional seismic series exist in a four-millennium catalog of $M \geq 6.5$ earthquakes, using an adapted η metric with tectonic distance along Bird (2003) plate boundaries. **Novelty.** To our knowledge, this is the first global application of nearest-neighbor clustering with tectonic-path distance across historical, early instrumental, and modern catalogs, combined with ETAS validation and FDR correction.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records from ~2150 BCE). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. USGS ComCat contains 2,088 raw $M \geq 6.5$ events (1973–2026); after merging and deduplication, 2,041 $M \geq 6.5$ events remain (no `quality_score` filter). Gardner–Knopoff declustering removes ~24 aftershocks (2,017 independent, 98.8 %). `quality_score` 0.30–0.95 is interpretive metadata (Woessner & Wiemer 2005), not an inclusion filter. Final catalog: 4,267 $M \geq 6.5$ events (4,418 CSV rows): 2,041 modern, 2,179 early instrumental, 47 historical (fragmentary paleoseismic).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b -value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

2.2 Tectonic distance and connectivity metric

Tectonic distance r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX). If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, $r_{ij} = 1.5 \times r_{GC}$ (great-circle Haversine).

$$\eta_{ij} = t_{ij} \cdot r_{ij}^{1.6} \cdot 10^{(-b \cdot m_i)}$$

Baiesi & Paczuski [2004]; $d_f=1.6$; $b=1.0$ (code default); parent m_i only. η is a relative measure; η_0 is empirical from the NN distribution.

2.3 Threshold η_0 and series detection

Automatic η_0 from nearest-neighbor η distribution: KDE valley between bimodal $\log_{10}(\eta)$ modes (Zaliapin & Ben-Zion 2013); default $\eta_0 = 10^{\text{median } \log_{10} \eta}$. Validated against Poisson permutation null ($n = 10,000$).

- 1. Declustering via Gardner–Knopoff [1974].
- 2. Nearest-neighbor forest: parent $i^* = \text{argmin } \eta_{ij}$ subject to $\eta_{ij} < \eta_0$.
- 3. Global series: $N \geq 4$; $M \geq 6.5$; ≥ 3 Flinn–Engdahl regions.
- 4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

2.4 Statistical validation

Permutation test: $n = 10,000$, $p < 0.0001$, $z = -6.17$ (modern). ETAS validation (ETASCatalogGenerator): $\mu=0.008$, $K=0.08$, $\alpha=1.0$, $c=0.005$ days, $p=1.1$; $\text{max_trigger_distance_km}=500$ (links >500 km forbidden). 100 synthetic catalogs (branching Poisson, seed=42); same global_series algorithm on each. FPR = fraction with ≥ 1 false series = 0/100; $p_{\text{ETAS}} = 0.0000$ (discrete test, resolution 1/101). FDR (Benjamini–Hochberg, $q = 0.05$): 45/47 series significant. Declustering: Gardner–Knopoff 98.8%; Zaliapin–Ben-Zion 100.0% independent.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 global seismic series: 27 modern ($p < 0.0001$), 15 early instrumental ($p = 0.007$; pre-1960 incompleteness caveat), 5 historical candidates (not significant, $p = 0.46$; 47 $M \geq 6.5$ events pre-1900). 142 cluster candidates before filtering.

ID	N	Regions	M _{max}	Period	Notes
S047	53	5	8.0	1982–2024	Western Pacific subduction corridor
S170	46	12	9.1	2002–2023	Sunda belt; Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	Western Pacific arc
S116	22	5	8.2	1993–2021	South Pacific multi-arc series
S191	15	4	8.4	2007–2022	Post-2004 activation episode

Table 1. Top five multi-regional series (≥ 3 Flinn–Engdahl regions). Fig. 1: figures/grl/fig01_global_map.png; Fig. 2: fig02_etas_validation.png.

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic distance lowers the η_0 threshold by $\sim 0.3 \log_{10}$ units compared with Euclidean distance.

4. DISCUSSION AND CONCLUSIONS

Observed global series are incompatible with Poisson and local-only ETAS nulls. Michael [2011] and Shearer & Stark [2012] tested global event rates, not multi-regional linkage structure; our η -metric is complementary. Limitations: historical $p=0.46$ (47 $M \geq 6.5$ events pre-1900); $p_{\text{ETAS}}=0.0000$ is a discrete 100-catalog test (does not prove mechanism); correlative η -metric without depth or focal mechanisms.

1. 4,267 $M \geq 6.5$ events (4,418 CSV records) contain 47 global series; 27 modern significant at $p < 0.0001$.
2. ETAS validation ($\mu=0.008$, $K=0.08$, $\alpha=1.0$, $c=0.005$ d, $p=1.1$, 500 km; FPR=0/100) and FDR (45/47) confirm non-randomness.
3. S170 (46 events, 12 regions, 2002–2023, $M_{\text{max}}=9.1$) demonstrates Pacific Ring activation.
4. Tectonic distance increases η sensitivity by $\sim 0.3 \log_{10}$.
5. Interpretive fork: (a) if series are real — hazard implications and long-range ETAS kernels; (b) if artifacts — FDR+ETAS remains a reproducible null-test pipeline.

Future work: ΔCFS for S047, S170, S095 with depth and focal mechanisms (King et al. 1994; Stein 1999); ETAS long-range kernels; Zenodo DOI release.

DATA AND CODE AVAILABILITY

Interactive presentation: marshalkin-ux.github.io/paleoseismic-clustering/ Source code: github.com/marshalkin-ux/paleoseismic-clustering. Results: [results/analysis_full_historical.json](#), [results/etas_validation.json](#), [results/fdr_correction_results.csv](#).

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